Question Bank

23INMCA304 - Fundamentals of Data Science

Course Type	Course Nature	CA Conduct	System	L	T	P	Credits	CA Total	CA Pass	SEE Total	SEE Pass	Total Pass
Theory	1	End Semester	Mark	3	1	0	4	40	24	60	24	50

Question Bank Summary

Sect. Part A	Sect. Part B	Easy	Med.	Chall.	Th.	Appli.
37	35	39	22	11	0	72

Part A

#	Unit	Question	COS	Categorized
1	1.1	A retail enterprise integrates ERP sales data with customer opinions collected from social media platforms. Analyze how differences in data structure influence the reliability and depth of business insights.	CO1	Medium - Analysing - A
2	1.1	A logistics company deploys GPS-enabled vehicles and RFID-tagged inventory for real-time tracking. Infer the data-related factors contributing to processing delays despite advanced infrastructure.	CO1	Medium - Analysing - A
3	1.1	An e-commerce platform maintains structured transaction records along with unstructured customer reviews. Analyze the reasons unified analytics across these datasets produces inconsistent outcomes.	CO1	Easy - Analysing - A
4	1.1	A banking system processes millions of online transactions every minute to prevent fraud. Analyze the relationship between data velocity and system performance degradation during peak hours.	CO1	Challenging - Analysing - A
5	1.1	A social media organization manages user-generated text, images, audio, and video content. Classify the analytical challenges introduced by such data diversity and infer their impact on decision-making.	CO1	Medium - Analysing - A
6	1.1	A healthcare provider integrates electronic health records with medical imaging and live sensor data. Analyze why analytical outcomes differ across departments using the same data platform.	CO1	Challenging - Analysing - A
7	1.1	An enterprise migrates from centralized storage to a Hadoop-based distributed environment. Analyze the operational changes that contribute to uneven efficiency gains across business units.	CO1	Easy - Analysing - A
8	1.1	A digital marketing firm relies on customer feedback containing slang, abbreviations, and inconsistencies. Infer how data uncertainty influences strategic decision stability.	CO1	Easy - Analysing - A
9	1.1	An online learning platform studies historical student performance and engagement duration. Analyze how patterns derived from past data support academic intervention strategies.	CO1	Easy - Analysing - A
10	1.1	A weather forecasting system continuously ingests satellite feeds and sensor readings. Analyze how data flow rate shapes system responsiveness and forecast accuracy.	CO1	Medium - Analysing - A
11	1.1	An organization merges internal operational data with external market intelligence. Analyze the role of preprocessing in improving the usefulness of combined datasets.	CO1	Easy - Analysing - A

12	1.1	A transportation authority applies large-scale analytics using mobile and roadside sensors. Analyze the factors that enable gradual improvement in traffic management decisions.	CO1	Challenging - Analysing - A
13	1.1	An organization accumulates vast datasets yet fails to derive actionable insights. Critically evaluate the gap between data availability and analytical value creation.	CO1	Challenging - Evaluating - A
14	2.1	A Big Data platform stores raw external data first and stores the processed data again after wrangling. Analyze how this storage approach impacts scalability and storage efficiency.	CO2	Easy - Analysing - A
15	2.1	A cluster consists of multiple nodes with identical hardware working as a single system. Analyze how task decomposition across nodes affects performance and fault tolerance.	CO2	Medium - Analysing - A
16	2.1	A distributed file system is used to store large streaming datasets rather than many small files. Analyze why file size influences access performance in distributed environments.	CO2	Easy - Analysing - A
17	2.1	A database system horizontally partitions data across multiple servers using sharding. Analyze how query patterns influence shard design and system efficiency.	CO2	Easy - Analysing - A
18	2.1	A system experiences partial data unavailability when a single node fails. Analyze how sharding contributes to partial fault tolerance in such scenarios.	CO2	Medium - Analysing - A
19	2.1	A content delivery platform implements master–slave replication to support millions of read requests per second. Analyze why read scalability improves significantly while write operations become a performance bottleneck.	CO2	Medium - Analysing - A
20	2.1	A large-scale system prioritizes service availability even when network partitions occur. Analyze how this design choice affects consistency expectations in the context of CAP theorem.	CO2	Medium - Analysing - A
21	2.1	A transactional system enforces atomicity and isolation across concurrent operations in a distributed environment. Analyze how these guarantees influence system throughput under heavy workloads.	CO2	Challenging - Analysing - A
22	2.1	A BASE-oriented database returns responses immediately after writes without ensuring global consistency. Analyze how eventual consistency enables system responsiveness at scale.	CO2	Easy - Analysing - A
23	2.1	A relational database relies on application-level logic to manage horizontal partitioning. Analyze how this design complicates cross-shard queries and transactional coordination.	CO2	Challenging - Analysing - A
24	2.1	A NoSQL database ingests semi-structured data with evolving formats at high speed. Analyze how schema-on-read supports adaptability in such ingestion pipelines.	CO2	Easy - Analysing - A
25	2.1	A distributed database allows all nodes to accept read and write operations under a peer-to-peer replication model. Analyze how conflict resolution mechanisms influence data consistency guarantees.	CO2	Easy - Analysing - A
26	3.1	A retail organization initiates a data analytics project after noticing declining customer retention but proceeds without clearly articulating the business objectives or success criteria. As the project progresses, technical insights emerge, yet decision-makers struggle to act on them. Analyze how weaknesses in the discovery phase influence alignment between analytics outcomes and business decisions.	CO3	Challenging - Analysing - A
27	3.1	During an analytics initiative, a data science team repeatedly revisits earlier lifecycle stages as new data patterns emerge and assumptions change. While this delays timelines, insight quality improves. Analyze how the iterative nature of the analytics lifecycle supports deeper problem understanding in complex business environments.	CO3	Easy - Analysing - A
28	3.1	An organization begins large-scale data acquisition from multiple sources before assessing relevance, completeness, and quality. Later phases reveal that significant portions of data are unusable. Analyze how inadequate data evaluation during early lifecycle stages affects modeling reliability and project cost.	CO3	Easy - Analysing - A

29	3.1	A data analytics team maintains detailed documentation for assumptions, transformations, and analytical decisions throughout the project lifecycle. New team members quickly adapt, and results remain reproducible. Analyze how lifecycle documentation strengthens analytical consistency and organizational learning.	CO3	Easy - Analysing - A
30	3.1	During data preparation, analysts use an isolated analytics sandbox to explore transformations, feature creation, and sampling strategies without affecting production systems. Analyze how sandbox-based experimentation enhances analytical flexibility while reducing operational risk.	CO3	Easy - Analysing - A
31	3.1	A team performs extensive extraction, transformation, and loading operations to standardize data collected from heterogeneous enterprise systems. Minor transformation errors later propagate into analytical models. Analyze how transformation decisions during data preparation shape downstream model accuracy and interpretability.	CO3	Easy - Analysing - A
32	3.1	While planning models, analysts explore correlations, distributions, and variable relationships before finalizing analytical techniques. Some initially preferred models are later discarded. Analyze how early data exploration influences the appropriateness and effectiveness of model selection.	CO3	Medium - Analysing - A
33	3.1	An analytics project separates datasets into training, validation, testing, and production environments. Performance metrics differ significantly across these stages. Analyze how environment separation contributes to realistic performance assessment and deployment readiness.	CO3	Easy - Analysing - A
34	3.1	A project sponsor evaluates analytical outcomes against predefined business metrics rather than purely statistical accuracy. As a result, some technically strong models are rejected. Analyze how business success criteria shape interpretation of analytical results.	CO3	Medium - Analysing - A
35	3.1	Before organization-wide rollout, an analytics solution is deployed as a pilot in a limited operational setting. Several unanticipated issues are identified and corrected. Analyze how pilot deployment reduces operational risk during the final lifecycle stage.	CO3	Easy - Analysing - A
36	3.1	An enterprise analyzes historical batch data for trend discovery while simultaneously processing real-time streams for operational alerts. Analyze how time sensitivity of data influences analytical technique selection across lifecycle phases.	CO3	Challenging - Analysing - A
37	3.1	A cross-functional analytics project involves business stakeholders, data engineers, and data scientists working collaboratively from discovery to deployment. Analyze how interdisciplinary collaboration impacts analytical relevance and adoption.	CO3	Easy - Analysing - A

Part B

#	Unit	Question	COS	Categorized
1	1.1	A multinational retail corporation accumulates transactional data, customer feedback, CCTV footage, and web interaction logs. Evaluate how data volume, variety, and velocity collectively influence architectural decisions for analytics platforms.	CO1	Challenging - Evaluating - A
2	1.1	A smart city initiative integrates traffic sensors, environmental monitors, surveillance systems, and citizen-generated social media data. Evaluate how interactions among Big Data elements shape effective urban governance mechanisms.	CO1	Easy - Evaluating - A
3	1.1	A financial institution continues to rely on relational databases for near real-time analytics on rapidly expanding datasets. Judge the suitability of this approach under current data conditions using clear technical criteria.	CO1	Easy - Evaluating - A
4	1.1	A hospital correlates historical clinical records with continuous health-monitoring streams to anticipate patient readmissions. Evaluate how this analytical shift enhances clinical decision-making quality.	CO1	Medium - Evaluating - A
5	1.1	An airline analyzes booking history, travel frequency, and passenger preferences to optimize services. Evaluate how data-driven insights simultaneously influence customer satisfaction and profitability.	CO1	Medium - Evaluating - A
6	1.1	A government agency combines census data with public sentiment extracted from social media platforms. Evaluate the limitations and risks associated with such data fusion in policy formulation.	CO1	Easy - Evaluating - A

7	1.1	An enterprise processes relational tables, JSON documents, emails, images, and video streams in a unified analytics system. Evaluate the structural factors that lead to variations in processing efficiency across data forms.	CO1	Easy - Evaluating - A
8	1.1	A telecom provider aims to reduce customer churn using usage logs, billing records, and network performance data. Evaluate how multiple analytics paradigms jointly support strategic churn management decisions.	CO1	Easy - Evaluating - A
9	1.1	A manufacturing unit relies on high-frequency sensor streams to anticipate equipment failures. Evaluate how real-time analytics transforms maintenance strategies from reactive to proactive.	CO1	Medium - Evaluating - A
10	1.1	A university adopts learning analytics to personalize instruction and assessment strategies. Evaluate the long-term academic implications of analytics-driven personalization.	CO1	Easy - Evaluating - A
11	1.1	A business plans enterprise-wide adoption of Big Data infrastructure across departments. Develop an integrated analytics vision that aligns cross-functional data usage with long-term competitiveness.	CO1	Medium - Creating - A
12	2.1	An enterprise processes massive volumes of unstructured logs, sensor streams, and user-generated content. Evaluate the effectiveness of distributed file systems in supporting scalable storage and parallel analytics compared to traditional databases.	CO2	Medium - Evaluating - A
13	2.1	A Big Data platform combines sharding for scalability and replication for fault tolerance across a cluster. Evaluate how this combination influences data availability, recovery behavior, and operational complexity.	CO2	Easy - Evaluating - A
14	2.1	A financial application requires uninterrupted service while maintaining strong data correctness guarantees. Evaluate the feasibility of meeting these requirements under CAP theorem constraints during network failures.	CO2	Medium - Evaluating - A
15	2.1	An organization migrates from a strictly ACID-compliant relational database to a BASE-oriented NoSQL system. Evaluate how this shift alters transactional reliability, latency, and application design assumptions.	CO2	Easy - Evaluating - A
16	2.1	A real-time IoT analytics platform generates high-velocity writes that overwhelm traditional relational databases. Evaluate how NoSQL data models address ingestion scalability and latency constraints.	CO2	Medium - Evaluating - A
17	2.1	An enterprise adopts a polyglot persistence strategy using relational, document, and key–value databases. Evaluate how this approach aligns storage technologies with workload diversity.	CO2	Easy - Evaluating - A
18	2.1	A recommendation engine relies on complex relationships among users, products, and interactions. Evaluate the suitability of graph databases for supporting such relationship-centric queries.	CO2	Medium - Evaluating - A
19	2.1	A real-time decision-support system demands extremely low latency and continuous data availability. Evaluate the role of in-memory data grids in meeting these operational demands.	CO2	Easy - Evaluating - A
20	2.1	An organization integrates an in-memory data grid with persistent disk storage using write-behind mechanisms. Evaluate the implications of this design on data durability, recovery time, and consistency.	CO2	Easy - Evaluating - A
21	2.1	An enterprise replaces disk-based relational databases with relational in-memory systems for OLTP workloads. Evaluate the trade-offs between performance gains and scalability limitations.	CO2	Challenging - Evaluating - A
22	2.1	A large organization supports transactional systems, batch analytics, and real-time processing simultaneously. Develop a cohesive storage architecture that integrates DFS, NoSQL, and in-memory technologies.	CO2	Medium - Creating - A
23	2.1	A cloud-based Big Data platform must dynamically scale resources while controlling operational costs. Design a storage strategy that balances performance, availability, fault tolerance, and cost efficiency.	CO2	Easy - Creating - A
24	3.1	An organization formally adopts the Data Analytics Lifecycle to manage complex Big Data initiatives across departments. While governance improves, teams express concern about reduced flexibility during exploration. Evaluate how lifecycle governance balances analytical discipline with innovation and experimentation.	CO3	Medium - Evaluating - A

25	3.1	In a data-driven project, business leaders prioritize rapid delivery of insights, whereas analytics teams advocate for extended discovery and preparation phases. Tension arises between speed and rigor. Evaluate how such competing priorities influence long-term project success across lifecycle stages.	CO3	Challenging - Evaluating - A
26	3.1	A company bases strategic decisions solely on numerical indicators while overlooking qualitative signals such as customer sentiment and behavioral feedback. Over time, market responsiveness declines. Evaluate the analytical limitations introduced by excluding qualitative insights from the analytics lifecycle.	CO3	Easy - Evaluating - A
27	3.1	A retail enterprise uses automated data mining techniques to uncover hidden patterns within massive customer datasets. Business teams rely heavily on these outputs for planning. Evaluate how automated discovery supports predictive decision-making while introducing interpretability challenges.	CO3	Medium - Evaluating - A
28	3.1	A marketing department observes strong correlations between promotional campaigns and sales growth and proceeds with aggressive investment. Later analysis reveals misleading relationships. Evaluate the risks associated with misinterpreting correlation as causation in analytical decision-making.	CO3	Easy - Evaluating - A
29	3.1	A retailer applies regression-based forecasting models to estimate product demand under varying environmental conditions. Strategic inventory decisions depend on these forecasts. Evaluate how regression modeling enhances predictive planning compared to purely descriptive analysis.	CO3	Easy - Evaluating - A
30	3.1	A financial institution applies supervised learning techniques to classify loan applicants by risk level. Model outputs directly influence approval decisions. Evaluate how training data quality and labeling accuracy influence classification reliability.	CO3	Easy - Evaluating - A
31	3.1	An organization clusters customers based on purchasing patterns to personalize marketing strategies. Some segments respond strongly, while others show limited engagement. Evaluate how unsupervised learning outcomes depend on feature selection and distance measures.	CO3	Easy - Evaluating - A
32	3.1	A fraud detection system relies on anomaly detection techniques to flag unusual transaction behavior. False positives create operational overhead. Evaluate the effectiveness of outlier detection in balancing risk reduction and operational efficiency.	CO3	Easy - Evaluating - A
33	3.1	A recommender system combines collaborative filtering with content-based techniques to improve personalization. User satisfaction improves but system complexity increases. Evaluate how hybrid recommendation strategies balance accuracy, scalability, and maintainability.	CO3	Easy - Evaluating - A
34	3.1	A customer experience platform analyzes call transcripts, chat logs, and survey responses using natural language processing and sentiment analysis. Insights must support strategic service improvements. Develop an integrated semantic analytics approach that aligns unstructured data insights with business objectives.	CO3	Medium - Creating - A
35	3.1	A smart city initiative combines time-series data, spatial information, and visual analytics dashboards for urban planning and forecasting. Decision-makers require both real-time and strategic insights. Design a comprehensive visual analytics framework that supports multi-scale decision-making.	CO3	Easy - Creating - A